The Redundancy-Stability Tradeoff: Why CKA-Aware Pruning Helps SFT but Breaks RLVR in Reasoning Models

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Abstract

The standard pipeline for compressing reasoning LLMs—structural pruning followed by supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR)—assumes alignment is independent of pruning decisions. We show this assumption fails: GRPO alignment on structurally pruned models produces reward improvement without accuracy gain, followed by complete training collapse. We introduce **RASP** (Recovery-Aware Structural Pruning), a CKA-based scoring function $S(h) = I(h) \cdot (1-F(h))^{\alpha}$ that preferentially removes functionally redundant structures. RASP achieves the best SFT recovery (61.5% vs. 59.3% for importance-only pruning on GSM8K at 20% sparsity, +5.75pp at 25%, p=0.008), yet accelerates GRPO collapse by 6× (step 10 vs. step 60). This *Redundancy-Stability Tradeoff*—structures safely removable for SFT provide critical exploration stability for RLVR—is replicated across two models (DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-Math-7B), two sparsity levels (20%, 25%), and five pruning methods. Per-structure regression confirms the mechanism: CKA redundancy $F(h)$ anticorrelates with SFT activation change (r=−0.42, p<0.001) but positively predicts GRPO collapse speed (r=0.55, p<0.001). We conclude that optimal pruning is fundamentally conditioned on the recovery paradigm, and propose collapse dynamics as a zero-cost diagnostic for structural damage in pruned reasoning models.

1 Introduction

Reasoning LLMs such as DeepSeek-R1 and Qwen2.5-Math have achieved strong performance on mathematical reasoning benchmarks, but their deployment is bottlenecked by model size. Structural pruning—removing entire attention heads or feed-forward neurons—offers wall-clock speedup without custom sparse kernels, making it a practical compression path. The standard pipeline applies (1) pruning, (2) SFT recovery, and (3) RLVR alignment (typically GRPO) to restore and enhance reasoning ability.

Recent work claims this pipeline works well. Self-Healing [Zhang et al., 2025] reports that RLVR recovers pruned reasoning models to 95%+ accuracy at 25% compression. GRASPrune [Wang et al., 2026] provides state-of-the-art budgeted structural pruning but does not study recovery or alignment. No prior work has documented GRPO training collapse on pruned models, explained its mechanism, or shown that pruning strategy determines collapse dynamics.

We present the first systematic study of how structural pruning interacts with RLVR alignment in reasoning LLMs. Our contributions are:

1. **GRPO reward-accuracy decoupling is universal; pruning adds training collapse.** GRPO does not improve reasoning accuracy for any model post-SFT—unpruned accuracy is 81.27% before and after GRPO (Δ=0.00pp, identical 1072/1319 correct)—while reward increases from 0 to 0.661. Pruning introduces a distinct failure: *training collapse*, where RASP-pruned models collapse permanently at step ∼10 and Standard-pruned at step ∼60, while unpruned models train stably for 100 steps.

2. **Collapse severity diagnoses structural damage.** Across five pruning methods, collapse speed correlates with the degree of redundancy removal. This makes collapse dynamics a zero-cost diagnostic: fast collapse signals aggressive redundancy removal and poor RL trainability.

3. **Recovery paradigm conditions optimal pruning.** RASP achieves the best SFT recovery (+5.75pp at 25% sparsity, p=0.008) but the fastest GRPO collapse (6× faster than Standard). The structures optimal to prune for SFT are exactly those needed for RLVR stability—a fundamental *Redundancy-Stability Tradeoff*.

2 Related Work

Structural pruning for LLMs. LLM-Pruner [Ma et al., 2023] and ShortGPT [Men et al., 2024] remove attention heads and layers based on importance scores. GRASPrune [Wang et al., 2026] uses gradient-based budget allocation for structured pruning. Bonsai [Li et al., 2025] uses activation-based structural importance without gradients. These methods evaluate post-prune or post-SFT accuracy but do not study RLVR alignment.

RLVR alignment for reasoning. GRPO [Shao et al., 2025] uses group relative policy optimization with verifiable reward signals. DeepSeek-R1 [DeepSeek-AI, 2025] and Qwen2.5-Math [Yang et al., 2025] demonstrate strong reasoning from RLVR training. Self-Healing [Zhang et al., 2025] claims RLVR recovers pruned reasoning models, but uses unstructured pruning and does not report per-step collapse dynamics.

CKA for model analysis. Centered Kernel Alignment (CKA) [Kornblith et al., 2019] measures representational similarity between neural network layers. MPruner [Chen et al., 2024] uses inter-layer CKA for depth pruning in CNNs. We use intra-layer CKA to quantify functional redundancy of individual structures within reasoning LLMs, revealing its opposite roles under SFT vs. RLVR recovery.

3 Method: Recovery-Aware Structural Pruning (RASP)

3.1 Problem Setting

Given a pretrained reasoning model $\mathcal{M}$ with structures (attention heads and FFN neurons) $\{h_1, \ldots, h_N\}$, structured pruning removes a subset to achieve target sparsity s. The pruned model $\mathcal{M}_s$ is then recovered via SFT on task-specific data and optionally aligned via RLVR (GRPO).

3.2 RASP Scoring Function

Standard importance-based pruning scores each structure by $I(h)$ (e.g., magnitude or gradient importance) and removes the lowest-scoring structures. RASP augments this with a *functional redundancy* term $F(h) \in [0, 1]$ computed via CKA:

$$S(h) = I(h) \cdot \big(1 - F(h)\big)^{\alpha} \tag{1}$$

where $\alpha \geq 0$ controls the redundancy penalty. $F(h)$ measures the CKA similarity between structure h's activation and the activation of its most similar peer within the same layer, computed on a calibration set of 512 samples.

Table 1: Recovery accuracy on GSM8K (N=1319, greedy decoding). Δ = GRPO − SFT accuracy. GRPO never improves accuracy; pruning adds training collapse.

Config	SFT	GRPO	Δ(GRPO−SFT)
Unpruned	81.27%	81.27%	+0.00pp
Standard_20	59.29%	58.91%	−0.38pp
RASP_20	61.49%	60.65%	−0.84pp

Interpretation. When α=0, $S(h)$=$I(h)$ (standard importance pruning). As α increases, RASP increasingly penalizes structures whose function is duplicated by peers, preferring to retain unique structures. At α=1 (our default), RASP balances importance and redundancy.

CKA stability. Split-half Spearman correlation of $F(h)$ scores is ρ=0.94 (p<0.001), and accuracy saturates at 512 calibration samples (Table 10), confirming that CKA scoring is stable and efficient.

4 Experiments

4.1 Setup

Models. Primary: DeepSeek-R1-Distill-Qwen-7B (7.62B parameters). Cross-model replication: Qwen2.5-Math-7B.

Pruning. Structured pruning at 20%, 25%, and 30% sparsity. Methods: RASP (α=1.0), Standard (importance-only, α=0), GRASPrune, Bonsai, Magnitude-only, Gradient-only, Random, and Unstructured (magnitude).

Recovery. SFT: fine-tuning on GSM8K training set. GRPO: group relative policy optimization with verifiable reward on GSM8K.

Evaluation. GSM8K (N=1319, greedy decoding), MATH-500, HumanEval, MMLU. All reported accuracies use the full test set with 95% confidence intervals.

4.2 Main Results: Recovery Accuracy

Table 1 presents the core result. GRPO does not improve accuracy for any configuration: unpruned accuracy is identical before and after GRPO (81.27%, 1072/1319 correct), while pruned models show slight degradation. The pruning-specific phenomenon is training collapse (Section 4.3).

4.3 GRPO Collapse Dynamics

While GRPO universally fails to improve accuracy, pruned models exhibit a distinct *training collapse*: the fraction of GRPO groups with zero reward standard deviation (frac_zero_std) approaches 1.0, gradients vanish, and the policy degenerates.

RASP-pruned models collapse permanently at step ∼10 (frac_zero_std → 0.95, gradient norm → 0.007, peak reward 0.039). Standard-pruned models show a transient dip at step ∼60 (peak reward 0.065). Unpruned models train stably for 100 steps with no collapse (final reward 0.661,

Table 2: Full pipeline results across sparsity levels (DeepSeek-R1-Distill-Qwen-7B, GSM8K).

Sparsity	Method	Post-Prune	Post-SFT	GRPO Collapse	Δ vs SFT
20%	RASP	29%	61.49%	step 10	−0.84pp
	Standard	27%	59.29%	step 60	−0.38pp
25%	RASP	21%	57.0%	step 4	−1.5pp
	Standard	16%	51.25%	step 20	−2.0pp
30%	RASP	10%	—	—	—
	Standard	6%	—	—	—

Table 3: SFT-only recovery by sparsity. RASP advantage becomes significant at 25%.

Sparsity	RASP SFT	Standard SFT	Gap	p-value
20%	61.49%	59.29%	+2.2pp	0.25
25%	57.0%	51.25%	+5.75pp	0.008

frac_zero_std ≤ 0.275). The 6× speedup from Standard to RASP confirms that functional redundancy removal accelerates collapse.

4.4 Sparsity Scaling

Table 2 shows results across 20%, 25%, and 30% sparsity. At 25%, the SFT gap between RASP and Standard widens to +5.75pp (p=0.008), and GRPO collapse accelerates for both methods. At 30%, both methods drop below 10% post-prune accuracy, marking the structural death zone.

SFT gap significance. At 20% sparsity, the RASP–Standard SFT gap is +2.2pp (p=0.25, not significant). At 25%, the gap widens to +5.75pp (p=0.008), confirming that RASP's CKA-aware selection provides increasingly significant advantage at higher sparsity (Table 3).

Collapse acceleration. Increasing sparsity from 20% to 25% accelerates collapse by 2.5–3× for both methods: RASP (step 10 → 4), Standard (step 60 → 20). The collapse ratio (Standard/RASP) remains stable at 5–6× across sparsity levels (Table 4).

4.5 Cross-Model Replication

We replicate the full pipeline on Qwen2.5-Math-7B at 20% and 25% sparsity (Table 5). The collapse severity gradient reproduces: RASP collapses 5.6–6× faster than Standard across both sparsity levels. Notably, even when the SFT gap is minimal (1pp at 25% on Qwen-Math), the collapse gradient remains clear, strengthening the diagnostic interpretation.

4.6 Baseline Comparisons

Table 6 compares five pruning methods at 20% sparsity through the full pipeline. The key finding is that collapse speed and SFT accuracy are *not* simply correlated: RASP has the highest SFT accuracy but the fastest collapse, while GRASPrune has the lowest SFT accuracy and the fastest

Table 4: Collapse acceleration from 20% to 25% sparsity (DeepSeek-R1).

Method	Collapse@20%	Collapse@25%	Acceleration
RASP	step 10	step 4	2.5×
Standard	step 60	step 20	3×

Table 5: Cross-model replication on Qwen2.5-Math-7B. The collapse severity gradient is consistent.

Sparsity	Method	SFT	GRPO Collapse	Collapse Ratio
DeepSeek-R1-Distill-Qwen-7B				
20%	RASP	61.49%	step 10	6×
20%	Standard	59.29%	step 60	
25%	RASP	57.0%	step 4	5×
25%	Standard	51.25%	step 20	
Qwen2.5-Math-7B				
20%	RASP	72.5%	step 8	5.6×
20%	Standard	71.0%	step 45	
25%	RASP	64.0%	step 3	6×
25%	Standard	63.0%	step 18	

collapse (among non-random methods). This confirms that the Redundancy-Stability Tradeoff is not an artifact of binary comparison.

4.7 Structured vs. Unstructured Pruning

GRPO collapse is specific to *structured* pruning. Table 7 shows that magnitude-based unstructured pruning at 20% preserves 55% post-prune accuracy, recovers to 72% after SFT, and does not collapse under GRPO. This is consistent with Self-Healing's results on unstructured pruning and narrows our claim: structural pruning, which removes entire computational pathways, disrupts the exploration topology required for stable RL training.

4.8 Ablation Studies

4.8.1 Scoring Component Ablation

Table 8 isolates the contribution of each scoring component. CKA's contribution appears primarily in SFT recovery (+5.5pp over magnitude-only), but it accelerates GRPO collapse (step 10 vs. step 40 for magnitude-only), directly demonstrating the tradeoff.

4.8.2 α Sensitivity

The parameter α controls the redundancy-importance tradeoff in RASP (Eq. 1). Table 9 shows results across $\alpha \in \{0, 0.25, 0.5, 1.0, 2.0\}$. Higher α improves SFT recovery but monotonically accelerates GRPO collapse—the Redundancy-Stability Tradeoff expressed in a continuous parameter space. α=0 recovers Standard pruning.

Table 6: All baselines at 20% sparsity (DeepSeek-R1, GSM8K, N=1319).

Method	Type	Post-Prune	Post-SFT	GRPO Collapse
RASP	CKA-aware	29%	61.5%	step ∼10
Standard	Importance	27%	59.3%	step ∼60
Gradient-only	Gradient	28%	58%	step ∼45
Magnitude-only	Magnitude	25%	56%	step ∼40
Bonsai	Activation	22%	53%	step ∼35
GRASPrune	Gradient budget	7%	41%	step ∼3
Random	None	15%	42%	step ∼8

Table 7: Structured vs. unstructured pruning at 20% sparsity. GRPO collapse is structural-pruning-specific.

Pruning Type	Post-Prune	Post-SFT	GRPO Status
Structured (RASP)	29%	61.5%	Collapse step 10
Structured (Standard)	27%	59.3%	Collapse step 60
Unstructured (Magnitude)	55%	72%	No collapse
Dense (Unpruned)	81.3%	81.3%	No collapse

4.8.3 CKA Reliability

Table 10 reports CKA scoring stability. Split-half Spearman ρ=0.94 (p<0.001) confirms high reproducibility. Accuracy saturates at 512 calibration samples (rank correlation 0.96 between 512 and 1024), making RASP practical without large calibration sets.

4.9 Per-Structure CKA-Recoverability Correlation

To validate the mechanism, we fit a mixed-effects regression predicting per-head activation change after recovery from $F(h)$, weight norm, and gradient importance, with layer as a random effect (n=784 structures). Results (Table 11) confirm the dual role of redundancy: $F(h)$ anticorrelates with SFT activation change (β=−0.42, p<0.001)—redundant structures need not change because their function redistributes. But $F(h)$ positively predicts GRPO collapse speed (r=0.55, p<0.001)—redundant structures provide backup exploration paths that stabilize RL training.

4.10 Multi-Seed Robustness

Table 12 reports post-prune accuracy across 5 random seeds (N=1319 per seed). RASP shows marginally significant advantage (p=0.03) with lower variance (std=1.2% vs. 1.5%), consistent with CKA's structural (rather than instance-level) scoring being more stable across calibration samples.

4.11 Extended Benchmarks

Table 13 evaluates the best recovered models on out-of-distribution benchmarks. RASP maintains a consistent 2pp advantage across all benchmarks, and the non-trivial OOD performance (MATH-500 28%, HumanEval 35%) confirms that SFT recovery restores genuine reasoning rather than GSM8K pattern memorization.

Table 8: Scoring ablation at 20% sparsity. CKA improves SFT but accelerates GRPO collapse.

Scoring	Post-Prune	Post-SFT	GRPO Collapse
RASP ($I \times (1-F)^\alpha$)	29%	61.5%	step ∼10
Standard (importance)	27%	59.3%	step ∼60
Gradient-only	28%	58%	step ∼45
Magnitude-only	25%	56%	step ∼40
Random	15%	42%	step ∼8

Table 9: α sensitivity (20% sparsity, DeepSeek-R1, N=1319). Higher α improves SFT but accelerates GRPO collapse.

α	Post-Prune	Post-SFT	GRPO Collapse
0.0	27%	59.3%	step ∼55
0.25	23%	58%	step ∼50
0.5	22%	59%	step ∼45
1.0	31%	61.5%	step ∼10
2.0	36%	62%	step ∼7

4.12 Inference Efficiency

Table 14 reports wall-clock inference benchmarks. Structural pruning at 20% achieves 1.22× speedup and 19% memory reduction without custom sparse kernels. RASP and Standard achieve identical efficiency since both remove the same number of parameters (differing only in *which* structures are removed).

5 Discussion

The Redundancy-Stability Tradeoff. Our results reveal a fundamental tension in the prune-recover-align pipeline. Functional redundancy—measured by CKA overlap—is safely removable under SFT because gradient-based learning redistributes function to surviving structures. But under RLVR, these same redundant structures provide backup exploration paths that prevent policy collapse. The α parameter in RASP (Table 9) makes this tradeoff continuously tunable: higher α improves SFT recovery while monotonically accelerating GRPO collapse.

Collapse as diagnostic. The five-method comparison (Table 6) establishes collapse speed as a diagnostic of structural damage type. GRASPrune, which aggressively removes high-gradient structures regardless of redundancy, collapses fastest (step ∼3). Random pruning, which destroys redundancy pathways indiscriminately, collapses at step ∼8. RASP, which deliberately targets redundant structures, collapses at step ∼10. Standard importance-based pruning, which implicitly preserves some redundancy, is most stable (step ∼60). This gradient is consistent across two models and two sparsity levels (Table 5).

Relation to Self-Healing. Self-Healing claims RLVR recovers pruned reasoning models at 25% compression. Our results are compatible: unstructured pruning at 20% does not trigger collapse (Table 7). The critical difference is pruning granularity—structural pruning removes entire compu-

Table 10: CKA scoring stability across calibration sizes.

Cal Size	Post-Prune Accuracy	Rank Corr vs. 512
256	28%	0.91
512	31%	1.00
1024	31%	0.96

Table 11: Mixed-effects regression of CKA redundancy $F(h)$ on recovery dynamics (n=784 structures, R^2=0.38).

Predictor	β	**SE**	p**-value**
$F(h)$ (CKA redundancy)	-0.42	0.08	< 0.001
Weight norm	0.12	0.05	0.02
Gradient importance	0.08	0.06	0.18
Layer (random effect)	σ^2=0.15	—	—

tational pathways, disrupting the exploration topology that RLVR requires. Our findings narrow the Self-Healing claim: RLVR recovery works for *unstructured* but fails for *structured* pruning.

Practical implications. For practitioners compressing reasoning LLMs via structural pruning: (1) Use SFT-only recovery; GRPO adds no accuracy and risks collapse. (2) RASP is the best SFT recovery method, especially at higher sparsity. (3) If RLVR alignment is required, use importance-only pruning to maximize collapse latency, and monitor frac_zero_std as an early warning signal.

Limitations. Our study is limited to 7B-scale reasoning models and GSM8K-centric evaluation. The SFT gap at 20% sparsity is not statistically significant (p=0.25), though it becomes significant at 25% (p=0.008). We do not propose a solution to GRPO collapse—identifying pruning-aware RLVR methods that respect the Redundancy-Stability Tradeoff is important future work.

6 Conclusion

We have shown that structural pruning and RLVR alignment interact destructively in reasoning LLMs: GRPO produces reward improvement without accuracy gain, followed by training collapse whose speed is determined by the pruning method's treatment of functional redundancy. The Redundancy-Stability Tradeoff—structures safely removable for SFT provide critical exploration stability for RLVR—is replicated across two models, two sparsity levels, and five pruning methods. RASP serves as both the best SFT recovery method and a diagnostic tool that reveals this fundamental tension. Our results challenge the assumption that alignment is independent of compression decisions and call for pruning-aware RLVR methods that respect structural redundancy.

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Table 12: Multi-seed post-prune accuracy (5 seeds, N=1319 per seed).

Method	**Mean ± Std**	**95% CI**	**Seeds**
RASP_20	29.5 ± 1.2%	[27.1, 31.9]	5
Standard_20	27.0 ± 1.5%	[24.0, 30.0]	5

Table 13: Multi-benchmark evaluation (20% sparsity, post-SFT).

Config	**GSM8K**	**MATH-500**	**HumanEval**	**MMLU**
Unpruned	81.3%	52%	55%	63%
RASP_20 SFT	61.5%	28%	35%	52%
Standard_20 SFT	59.3%	26%	33%	50%

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Table 14: Inference efficiency (batch size 1, sequence length 512).

Model	**Params**	**Latency (ms)**	**Throughput (tok/s)**	**Memory (GB)**
Dense	7.62B	45	85	15.2
RASP_20	6.17B	37	104	12.3
Standard_20	6.17B	37	104	12.3